Construcciones tipo Lebeau-Robbiano

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Seminario de Análisis Microlocal.

25 de abril de 2026

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A general time iteration

Lebeau-Robbiano strategy is a time-iteration argument which balances the cost of small-time null controllability of eigenprojections and its dissipation rate.

Assumption

There exists a family $(P_k)_{k\in\mathbb{N}}$ *of proyections on* Y *and there exists constantes* $C_1, C_2, c_1, c_2, a, b, m > 0$ *with* $a < b$ *such that, for all* $k \in \mathbb{N}$,

- **Control of low modes.** *For all* $T \in (0,1]$ *and* $y_0 \in Y$, *there exists* $u \in L^p(\mathbb{R}_+; U)$ *such that*

$$P_k \Sigma_T(y_0, u) = 0 \quad \text{and} \quad \|u\|_{L^p} \leq C_1 e^{c_1 k^a} \|y_0\|_Y, \tag{1}$$

- **Dissipation of high modes.** *For all* $T \in (0,1]$ *and* $y_0 \in Y$,

$$P_k y_0 = 0 \quad \Rightarrow \quad \|\Sigma_T(y_0, 0)\|_Y \leq C_2 e^{-c_2 T^m k^b} \|y_0\|_Y. \tag{2}$$

Schematic process

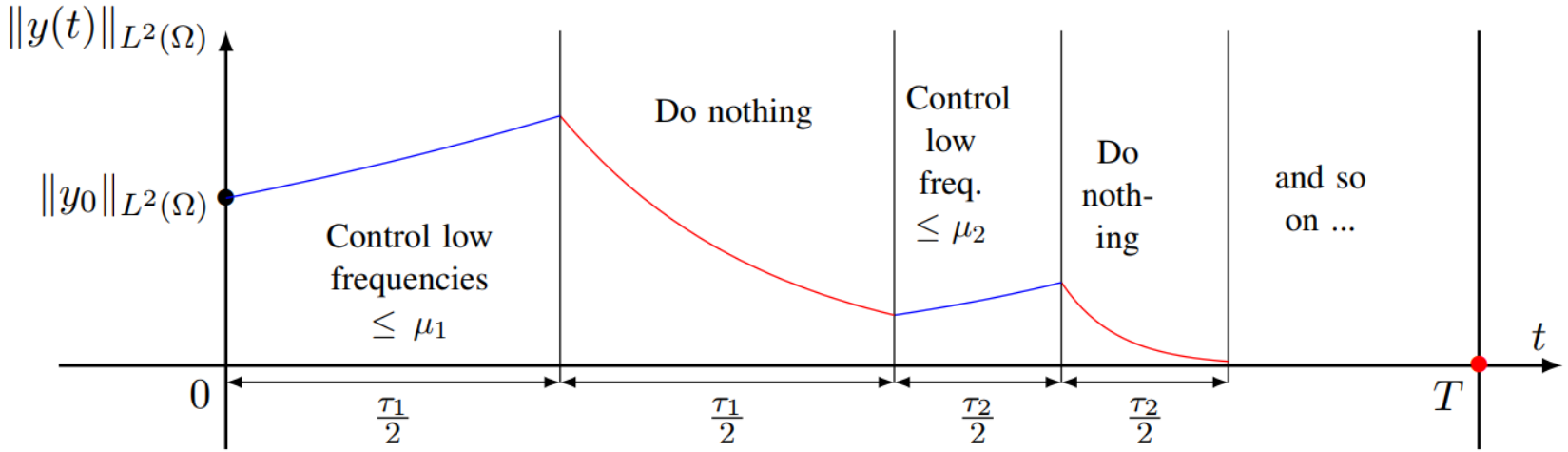


Figure IV.1: The Lebeau-Robbiano method

F. Boyer (2023)
Controllability of linear parabolic equations and systems.

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One-step contraction (OSC)

Idea: Partition time interval to apply one of the previous estimates. In each $\tau-$step, control low modes to zero on $[0, \tau/2]$ and let energy dissipate on $[\tau/2, \tau]$. The main time iteration relies on the following one-step contraction result:

Lemma

Suppose that Assumption 1.1 holds. Then there exists $C_3, c_3 > 0$ such that, for all $k \in \mathbb{N}$, for all $\tau \in (0, 1]$ and $y_0 \in Y$, there exists $u \in L^p(\mathbb{R}_+; U)$ such that

$$\|u\|_{L^p} \leq C_1 e^{c_1 k^a} \|y_0\|_Y, \tag{3}$$

$$\|\Sigma_\tau(y_0, u)\|_Y \leq C_3 e^{c_1 k^a} e^{-c_3 \tau^m k^b} \|y_0\|_Y. \tag{4}$$

Theorem

Theorem

Suppose that Assumption 1.1 holds. Then the abstract linear system Σ is small-time null-controllable. More precisely, there exists constants $C_0, c_0 > 0$ such that, for every $T \in (0,1]$ and $y_0 \in Y$, there exists $u \in L^p((0,T);U)$ such that $\Sigma_T(y_0,u) = 0$ and

$$\|u\|_{L^p} \leq C_0 \exp\left(c_0 T^{-\sigma}\right) \|y_0\|_Y, \qquad \sigma := \frac{am}{b-a}. \tag{5}$$

Example: interior control of the heat equation on [0,1].

$$\begin{cases} y_t - y_{xx} = u1_\omega, & (0,T)\times(0,1), \\ y(t,0) = y(t,1) = 0, & (0,T), \\ y(0,x) = y_0(x), & (0,1). \end{cases} \tag{6}$$

$y_0 \in L^2(0,1), u \in L^2((0,T)\times\omega)$ and $\omega \subset (0,1)$. The system is well-posed in

$$\mathcal{Y} := C^0([0,T]; L^2(0,1)) \cap L^2((0,T); H_0^1(0,1)) \cap H^1((0,T); H^{-1}(0,1)),$$

Moreover, there exists a constant $C > 0$ such that for every T, f, y_0,

$$\|y\|_{C^0L^2} + \|y\|_{L^2H^1} + \|y\|_{H^1H^{-1}} \leq C(\|y_0\|_{L^2} + \|f\|_{L^2H^{-1}}).$$

Energy dissipation and spectral inequality

$A = -\partial_{xx}$ with Dirichlet conditions admits the eigen-decomposition

$$\varphi_j(x) := \sqrt{2}\sin(j\pi x), \qquad \lambda_j = (j\pi)^2.$$

If $\langle y_0, \varphi_j \rangle = 0$ for $1 \leq j \leq k$, one has

$$\|y(t)\|_{L^2} \leq e^{-(k\pi)^2 t}\|y_0\|_{L^2}. \tag{7}$$

We assume the next spectral inequality due to Turán,

Proposition

Let $\omega \subset [0,1]$ be a measurable set with $|\omega| > 0$. there exists $C_\omega > 0$ such that for any $n \geq 1$ and $a_1, \ldots, a_n \in \mathbb{R}$,

$$\|f\|_{L^2(0,1)} \leq C_\omega^n \|f\|_{L^2(\omega)}, \quad f(x) := \sum_{j=1}^{n} a_j \sin(j\pi x).$$

Small-time null controllability

Theorem

Assume that $|\omega| > 0$. *Previous system is small-time null controllable and there exists* $C > 0$ *such that, for all* $T > 0$ *and* $y_0 \in L^2(0,1)$, *there exists* $u \in L^2((0,T) \times \omega)$ *such that the associated solution to* (6) *satisfies* $y(T) = 0$ *and* $\|u\|_{L^2} \leq ce^{C/T}\|y_0\|_{L^2}$.

Proof: Denote by P_k the projection on $\text{span}\{\varphi_1, \ldots, \varphi_k\}$ in $L^2(0,1)$. Energy dissipation of high modes holds with $b = 2$ and $m = 1$. Since P_k commutes with $\mathbb{T}_t$ and $B \in \mathcal{L}(L^2(\omega); L^2(0,1))$ is the extension by 0, so B^* is the restriction to ω. Hence by the spectral inequality, for any $y \in L^2(0,1)$,

$$\|P_k y\|_Y \leq e^{c_\omega k}\|B^* P_k y\|_U.$$

Thus we can apply the next Proposition

$$P_k \Sigma_T(y_0, u) = 0, \quad \text{and} \quad \|u\|_{L^p} \leq T^{1/p-1} M_1 C_1 e^{c_1 k^a}\|y_0\|_Y,$$

which implies that the control cost estimate holds with $a = 1$.

End of the proof

Thus Proposition 1.1 holds and Theorem 2 applies.
Since $p = 2$, the cost estimate from previous Proposition leads to a singular $T^{-1/2}$ prefactor, which is an absorbable weak degeneracy with the same iteration.

Todays session

Under an spectral inequality assumption, prove an estimate for the control when

1. Dynamics and spectral projections commute.
2. Dynamics and spectral projections do not commute.

Then we give an historical perspective and some applications.

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How to prove the previous assumption?

Thanks to Assumption 1.1, the Lebeau-Robbiano method can be applied as a black box. Many authors rely on spectral inequalities to check properties of control for low modes and dissipation for high ones.

We will suppose that U and Y are Hilbert spaces, $(P_k)_{k\in\mathbb{N}}$ is a family of orthogonal projections on Y and $\mathbb{T}_t \in \mathcal{L}(Y)$ is such that $\mathbb{T}_t y_0 := \Sigma_t(y_0, 0)$. Assume that there exists an operator $B \in \mathcal{L}(U, Y)$ such that for all $u \in L^p(\mathbb{R}_+; U)$ and $T \geq 0$,

$$\Sigma_T(0, u) = \int_0^T \mathbb{T}_{T-t} B u(t) dt.$$

General spectral inequality

Assumption (Spectral inequality)

There exists $C_1, c_1, a > 0$ *such that, for all* $k \in \mathbb{N}$,

$$\forall y \in Y, \quad \|P_k y\|_Y \leq C_1 e^{c_1 k^a} \|B^* P_k y\|_U, \tag{8}$$

or, equivalently, by the closed-graph theorem,

$$\exists D_k \in \mathcal{L}(Y, U), \text{ such that } P_k = P_k B D_k \text{ and } \|D_k\| \leq C_1 e^{c_1 k^a}. \tag{9}$$

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Commutative case

Proposition

Under Assumption 2.1, if the projections P_k commute with the dynamics, then, for all $k \in \mathbb{N}$, $T \in (0,1]$ and $y_0 \in Y$, there exists $u \in L^p((0,T);U)$ such that

$$P_k \Sigma_T(y_0, u) = 0, \quad \text{and} \quad \|u\|_{L^p} \leq T^{1/p-1} M_1 C_1 e^{c_1 k^a} \|y_0\|_Y, \tag{10}$$

where M_1 is the constant from Lemma **??**.

Proof: Suppose that 2.1 is valid. Let $k \in \mathbb{N}$ and $D_n \in \mathcal{L}(Y,U)$ be given by the Spectral Inequality. For $T \in (0,1]$ and $y_0 \in Y$, set

$$u(t) := -\frac{1}{T} D(k) T_t y_0. \tag{11}$$

By Lemma 1, we have the estimate

$$\|u\|_{L^p} \leq T^{1/p-1} \|D_k\| \cdot \sup_{t\in[0,1]} \|\mathbb{T}_t\| \cdot \|y_0\|_Y \leq T^{1/p-1} M_1 C_1 e^{c_1 k^a} \|y_0\|_Y. \tag{12}$$

Proof pt. 2

By linearity,

$$\Sigma_T(y_0, u) = \mathbb{T}_T y_0 - \frac{1}{T}\int_0^T \mathbb{T}_{T-t} B D_k \mathbb{T}_t y_0 dt. \tag{13}$$

By assumption, we know that $P_k \mathbb{T}_\tau P_k = P_k \mathbb{T}_\tau$ for every $\tau \geq 0$. Applying P_k to 13 and using $P_k B D_k = P_k$, the semigroup theory and the previous commutative property implies

$$\begin{aligned} P_k \Sigma_T(y_0, u) &= P_k \mathbb{T}_T y_0 - \frac{1}{T} P_k \int_0^T \mathbb{T}_{T-t} P_k B D_k \mathbb{T}_t y_0 dt \\ &= P_k \mathbb{T}_T y_0 - \frac{1}{T} P_k \int_0^T \mathbb{T}_{T-t} P_k \mathbb{T}_t y_0 dt \\ &= P_k \mathbb{T}_T y_0 - \frac{1}{T} P_k \int_0^T \mathbb{T}_{T-t} \mathbb{T}_t y_0 dt = 0. \end{aligned}$$

Non-commutative case

Suppose that $\mathbb{T}_t$ and P_k do not commute.

Proposition

Assume that there exist $C_1, C_2, c_1, c_2, a, b, m > 0$ with $a < b$ such that, for all $k \in \mathbb{N}$, the spectral inequality from Assumption 2.1 and the dissipation condition 2 hold. Then there exists $C_3, c_3 > 0$ such that, for all $k \in \mathbb{N}$, for all $k \in \mathbb{N}$, for all $\tau \in (0, 1]$ and $y_0 \in Y$, there exists $u \in L^p(\mathbb{R}_+; U)$ with

$$\|u\|_{L^p} \leq 2\tau^{1/p-1} M_1 C_1 e^{c_1 k^a} \|y_0\|_Y, \tag{14}$$

$$\|\Sigma_\tau(y_0, u)\|_Y \leq C_3 e^{c_1 k^a} e^{-c_3 \tau^m k^b} \|y_0\|_Y, \tag{15}$$

where M_1 is the constant from Lemma 1.

Proof pt. 1

Proof: Let $k \in \mathbb{N}$ and $D_k \in \mathcal{L}(Y, U)$ given by Proposition 2.1 on $(0, \tau/2)$, which is 0 on $(\tau/2, \tau)$:

$$u(t) := -\frac{2}{\tau}\mathbf{1}_{(0,\tau/2)}(t) D_k \mathbb{T}_t y_0. \tag{16}$$

By Lemma 1, we obtain the first estimate. By linearity,

$$\Sigma_\tau(y_0, u) = \mathbb{T}_t y_0 - \frac{2}{\tau}\int_0^{\tau/2} \mathbb{T}_{\tau - T} B D_k \mathbb{T}_t y_0 dt. \tag{17}$$

Then we write the following decomposition

$$BD_k = \mathrm{Id} + (BD_k - \mathrm{Id}) = \mathrm{Id} + P_k(BD_k - \mathrm{Id}) + (\mathrm{Id} - P_k)(BD_k - \mathrm{Id}). \tag{18}$$

By construction, $P_k(BD_k - \mathrm{Id}) = 0$. By linearity and the semigroup property,

$$\Sigma_\tau(y_0, u) = -\frac{2}{\tau}\int_0^{\tau/2} \mathbb{T}_{\tau - t}(\mathrm{Id} - P_k)(BD_k - \mathrm{Id})\mathbb{T}_t y_0 dt. \tag{19}$$

Proof pt. 2

Hence, by the dissipation condition 2,

$$\begin{aligned}\|\Sigma_\tau(y_0,u)\|_Y &\leq \frac{2}{\tau}\int_0^{\tau/2} \|\mathbb{T}_{\tau-t}(\mathrm{Id}-P_k)\|(\|B\|\|D_k\|+1)\|\mathbb{T}_t\|\|y_0\|_Y dt \\ &\leq C_2 e^{-c_2(\tau/2)^m k^b}(\|B\|C_1 e^{c_1 k^a}+1)M_1\|y_0\|_Y.\end{aligned}$$

This we obtain the last estimate with $C_3 = C_2(C_1\|B\|+1)M_1$ and $c_3 = c_2 2^{-m}$. □

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The 1995 paper of Lebeau and Robbiano

To prove the small-time null controllability of the heat equation posed on a smooth compact Riemannian manifold $\mathcal{M}$, the strategy is as follows.

1. Consider the family of eigenfunctions $(e_k)_{k\in\mathbb{N}}$ of the Dirichlet Laplacian $A = -\Delta$, with associated eigenvalues $(\lambda_k)_{k\in\mathbb{N}}$. Consider P_k to be the projection on the finite-dimensional space spanned by $e_1, \dots, e_k$.
2. The previous automatically ensures that $\|\Sigma_T(y_0, 0)\|_Y \leq e^{-\lambda_k T}\|y_0\|_Y$ when $P_k y_0 = 0$. By Weyl's law, this entails the dissipation property 2 with $b = 2/d$ and $m = 1$.
3. Using Carleman estimates for the elliptic operator $\partial_t^2 + \Delta$, prove that for any $\theta > 1$, for any $T \in (0, 1]$ and $y_0 \in Y$, there exists $u \in L^\infty(0, T)$ such that $P_k \Sigma_T(y_0, u) = 0$ with a cost estimate of the form

$$\|u\|_{L^\infty} \leq C_1 \exp\left(c_1 \lambda_k^{1/2} + c_1 T^{-\theta}\right) \|y_0\|_Y. \tag{20}$$

This corresponds to our generalized cost assumption with $a = 1/d$ by Weyl's law.

The 1995 paper of Lebeau and Robbiano

4. Use a geometric time and frequency schedule to deduce small-time null controllabiltiy of the full PDE. By Proposition ..., since $\theta > 1$ and $\frac{ma}{b-a} = 1$, their proof concludes to a cost estimate of the form $C_0 \exp(c_0 T^{-\theta})$ wiht $\theta > 1$, failing just short of the now-known optimal exponent 1.

More recent variations and enhancements

The Lebeau-Robbiano strategy has been applied in countless settings and many authors have proposed improvements.

- In 1999 Jerison, Lebeau and Zuazua started to isolate and study precisely static spectral inequalities of the form of Assumption 2.1, avoiding the lossy estimate about the control 5.
- In 2006 Miller started the first abstraction of the previous results which concerned the heat equation to self-adjoint operators A with admissible control operators B, obtaining the qualitatively optimal cost as $T \to 0$ for the control of a family of diffusion equations.
- Subsequent works improved the estimates of the cost of controllability or reduced the assumptions on the control operator B. Miller also remarked in 2010 that commutation is not necessary: we only need stability of the high modes as in Remark 2.14

More recent variations and enhacements

- Beauchard, Pravda-Starov and Miller in 2018 introduced the key novelty that projections P_k don't need to be spectral projections onto the eigenspaces of the operator A, which means that it is not required to be a self-adjoint operator. This important improve allows to reuse well-known spectral inequalities for a reference operator and apply it to prove the controllability of another PDE. See Kolmogorov example
- Later it was identified that the method can be extended to time-dependent systems (with time-dependent operators in 2017 by Beauchard and Pravda-Starov or time-dependent control domains in 2020 by Beauchard, Egidi and Pravda-Starov).

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Boundary control of the heat equation on $[0, \pi]$

Here we investigate for $T > 0$ the problem

$$\begin{cases} y_t - y_{xx} = 0, \quad (0, T) \times (0, \pi), \\ y(t, 0) = u(t), \quad (0, T), \\ y(t, \pi) = 0, \quad (0, T), \\ y(0, x) = y_0(x), \quad (0, \pi). \end{cases} \tag{21}$$

Given $y_0 \in H^{-1}(0, \pi)$ and $u \in L^2(0, T; \mathbb{R})$, there exists an unique weak solution $y \in C^0([0, T]; H^{-1}(0, \pi)) \cap L^2((0, T); L^2(0, \pi))$ defined by transposition. Denote by $\varphi_j(x) = \sin(jx)$ and $\lambda_j = j^2$ to the eigen-decomposition of $-\partial_{xx}$. We get

$$\frac{d}{dt}\langle y, \varphi_j \rangle = ju(t) - j^2 \langle y, \varphi_j \rangle,$$

so by Duhamel representation formula,

$$\langle y(T), \varphi_j \rangle = e^{-j^2 T} \langle y_0, \varphi_j \rangle + j\sqrt{2/\pi} \int_0^T e^{-j^2(T-t)} u(t) dt. \tag{22}$$

Energy dissipation: If $\langle y_0, \varphi_j \rangle = 0$ for all $1 \leq j \leq k$, one has

$$\|y(t)\|_{L^2} \leq e^{-k^2 t} \|y_0\|_{L^2}$$

.

Cost of control for low frequencies

To control low frequencies we rely on the Proposition 1.9, since there exists an $u \in L^2((0,T);\mathbb{R})$ such that $\langle y(T), \varphi_j \rangle = 0$ for all $1 \leq j \leq k$. It means that we have

$$\|u\|_{L^2} \leq \sqrt{\frac{k^2}{T}} \exp\left(\sqrt{\frac{k}{T}}\right) \|y_0\|_{H^{-1}}.$$

Using $x \leq e^x$ and Young inequality we obtain for $T \in (0,1]$,

$$\|u\|_{L^2} \leq e^{4(k-T^{-1})} \|y_0\|_{H^{-1}}.$$

Small-time null controllability

Typically the Lebeau-Robbiano strategy does not apply for boundary controllability since the spectral inequality required is not satisfied See references for boundary Carleman estimates. However, we can use the time iteration of the Lebeau-Robbiano method and another technique to achieve the controllability of the low frequencies.

Theorem

System 21 is small-time null controllable.

Demostración.

Let P_k be the projections on span$\{\varphi_1, \ldots, \varphi_k\}$. By energy dissipation, 2 holds with $b = 2$ and $m = 1$. By the previous estimate on u, the relaxed cost estimate **??** holds with $a = 1$ and $\sigma = 1$. Thus Assumption 1.1 holds and Theorem 2 applies. □

Control of the heat equation in $\mathbb{R}^d$ from a thick set

$$\begin{cases} y_t - \Delta y = u 1_\omega, & (0,T)\times\mathbb{R}^d, \\ y(0,x) = y_0(x), & \mathbb{R}^d. \end{cases} \tag{23}$$

Lemma

For any $T>0$, $y_0 \in L^2(\mathbb{R}^d)$ and $u \in L^1((0,T);L^2(\omega))$, there exists an unique solution $y \in C^0([0,T];L^2(\mathbb{R}^d))$ to system 23.

Energy dissipation and projections

In absence of control, the Fourier transform reads

$$\hat{y}(t,\xi) = e^{-t|\xi|^2}\hat{y}_0(\xi).$$

In particular, if $\operatorname{supp}\hat{y}_0 \subset B_{\mathbb{R}^d}(0,k)$ for some $k \geq 0$, one has

$$\|y(t)\|_{L^2} \leq e^{-tk^2}\|y_0\|_{L^2}.$$

Spectral inequality

We need to find an spectral inequality of the form

$$\|P_k y\|_{L^2(\mathbb{R}^d)} \leq C_k \|P_k y\|_{L^2(\omega)}.$$

For the system 23 to be small-time controlalble, it is natural to require that the control domain ω is well spread-out all over $\mathbb{R}^d$. To this, we use the next definition

Definition

Let $\gamma \in (0,1]$ and $a > 0$. We say that a measurable $\omega \subset \mathbb{R}^d$ is a $(\gamma, a)-$thick when

$$\forall x \in \mathbb{R}^d, \qquad \left|\omega \cap (x + [0,a]^d)\right| \geq \gamma a^d.$$

In this kind of sets we can apply the next equality.

Proposition

There exists an universal constant $C > 0$ such that for any $\gamma \in (0, 1]$, $a > 0$ and any measurable set $\omega \subset \mathbb{R}^d$ which is $(\gamma, a)-$thick, one has

$$\|f\|_{L^2(\mathbb{R}^d)} \leq \left(\frac{C}{\gamma}\right)^{C(N_a+1)} \|f\|_{L^2(\omega)},$$

for all $N \geq 0$ and $f \in L^2(\mathbb{R}^d; \mathbb{C})$ such that $\operatorname{supp}(\hat{f}) \subset B_{\mathbb{R}^d}(0, N)$.

Small-time null controllability:

Theorem

Let $\gamma \in (0, 1]$ and $a > 0$. Assume that ω is a (γ, a)-thick. Then the heat equation 23 is small-time null controllable.

Proof

Define P_k as the projection of the Fourier modes $|\xi| \leq k$. More precisely, set

$$P_k y := \mathcal{F}^{-1}(\xi \mapsto \hat{y}(\xi) 1_{\{|\xi| \leq k\}}).$$

We already have energy dissipation. Since ω is a (γ, a)-thick set, by the proposition we have for any $y \in L^2(\mathbb{R}^d)$,

$$\|P_k y\|_{L^2(\mathbb{R}^d)} \leq C_1 e^{c_1 k} \|P_k y\|_{L^2(\omega)},$$

for some constants $C_1, c_1 > 0$ depending on a, γ. We have that P_k commutes with the semigroup thanks to the Fourier solution. Then $B : L^2(\omega) \to L^2(\mathbb{R}^d)$ is the extension by 0 operator and B^* is merely the restriction to ω with $\|P_k y\|_{L^2(\omega)} = \|B^* P_k y\|_{L^2(\omega)}$. Thus by Proposition 3.1, for any $T \in (0, 1]$ and $y_0 \in L^2(\mathbb{R}^d)$, there exists $u \in L^1((0, T); L^2(\omega))$ such that $P_k \Sigma_T(y_0, u) = 0$ and $\|u\|_{L^1} \leq M_1 C_1 e^{c_1 k} \|y_0\|_{L^2(\mathbb{R}^d)}$. Hence 1 holds with $a = 1$.

Thus Assumption 1.1 holds and the result follows from Theorem 2.

Example: 2D Kolmorgorov equation

$$\begin{cases} y_t + zy_x - y_{zz} = u1_\omega, & (0,T)\times\mathbb{R}^2, \\ y(0,x,z) = y_0(x,z), & \mathbb{R}^2. \end{cases} \tag{24}$$

where $y_0 \in L^2(\mathbb{R}^2)$ is the initial data and $u \in L^1((0,T);L^2(\omega))$ is a control suspported on a measurable subset $\omega \subset \mathbb{R}^2$.

Proposition

For any $T > 0$, $y_0 \in L^2(\mathbb{R}^2)$ and $u \in L^1((0,T);L^2(\omega))$, there exist a unique solution $y \in C^0([0,T];L^2(\mathbb{R}^2))$ to 24.

We will use projection on Fourier modes instead of eigenspaces, as in the previous heat equation.

Energy dissipation

Proposition

There exists $c_0 := (4 - \sqrt{13}/6) > 0$ *such that, for any* $k \geq 1$ *and* $y_0 \in L^2(\mathbb{R}^2)$ *such that* $\hat{y_0} = 0$ *on* $B(0, k)$*, we have for any* $t \geq 0$*,*

$$\|y(t)\|_{L^2} \leq e^{-c_0 \min\{t, t^3\} k^2} \|y_0\|_{L^2}.$$

Proof: We perform a change on variables to follow the characteristics of the Kolmogorov equation. Set $\phi(t, x, z) := y(t, x - zt, z)$. In particular, $\phi(0) = y_0$. Then since

$$\partial_t y = \partial_t \phi - z\partial_x \phi, \quad \partial_x y = \partial_x \phi, \qquad \partial_{zz} y = \partial_{zz}\phi - 2t\partial_{xz}\phi + t^2 \partial_{xx}\phi,$$

we get

$$(\partial_t - \partial_{zz} + 2t\partial_{xz} - t^2\partial_{xx})\phi = 0.$$

Hence we have a PDE with time-varying coefficients which are constant in space. Taking Fourier transform in both directions $(x, z) \mapsto (\xi, \zeta)$ we get

$$(\partial_t + \zeta^2 + 2t\xi\zeta + t^2\xi^2)\hat{\phi} = 0.$$

Explicit integration leads to

$$\hat{\phi}(t, \xi, \zeta) = \exp\left(-t\zeta^2 - t^2\xi\zeta - \frac{t^3}{3}\xi^2\right)\hat{\phi}_0(\xi, \zeta).$$

Writing

$$|t^2\xi\zeta| \leq \frac{\alpha}{2}t\zeta^2 + \frac{1}{2\alpha}t^3\xi^2,$$

and optimizing with respect to $\alpha := (2 + \sqrt{13})/3$ we obtain that

$$t\zeta^2 + t^2\xi\zeta + \frac{t^3}{3}\xi^2 \geq \frac{4 - \sqrt{13}}{6}(t\zeta^2 + t^3\xi^2).$$

Since the initial data is supported for $\zeta^2 + \xi^2 \geq k^2$ we obtain that

$$\|\phi(t)\|_{L^2} \leq e^{-c_0 \min\{1, t^2\}k^2}\|\phi(0)\|_{L^2},$$

Small-time null controllability

Theorem

Let $\gamma \in (0,1]$ and $a > 0$. Asume that ω is (γ, a)-thick. Then the Kolmogorov equation 24 is small-time null controllable.

Proof: Define P_k as the projection on the Fourier modes $|\xi|^2 + |\zeta|^2 \leq k^2$.
First, by the previous estimate, the dissipation condition 2 holds with $m = 3$ and $b = 2$.
Second, since ω is a (γ, a)-thick, by Proposition **??**, fir any $y \in L^2(\mathbb{R}^2)$,

$$\|P_k y\|_{L^2(\mathbb{R}^2)} \leq C_1 e^{c_1 k} \|P_k y\|_{L^2(\omega)},$$

for some constants $C_1, c_1 > 0$ depending on a, γ. Hence $B : L^2(\omega) \to L^2(\mathbb{R}^2)$ is the extension by 0 and B^* the restriction to ω with $\|P_k y\|_{L^2(\omega)} = \|B^* P_k y\|_{L^2(\omega)}$. Note that P_k does not commute with $\mathbb{T}_t$, but we can apply the non-commutative case, then the One-step contraction applies and Theorem 2 ends the proof.